Laws of Exponents

Five rules let you combine powers without expanding them. Master when to add, subtract, or multiply the exponents — this is the heart of the chapter.

The five laws

In all of these the bases must be the same (for the first three) — that is the key condition.

1. **Product law — same base, ADD:** $n^a \times n^b = n^{a+b}$. e.g. $2^3 \times 2^5 = 2^8$.
2. **Quotient law — same base, SUBTRACT:** $n^a \div n^b = n^{a-b}$. e.g. $7^9 \div 7^4 = 7^5$.
3. **Power of a power — MULTIPLY:** $(n^a)^b = n^{a\times b}$. e.g. $(5^2)^3 = 5^6$.
4. **Power of a product:** $n^a \times m^a = (n \times m)^a$. e.g. $4^3 \times 5^3 = 20^3$.
5. **Power of a quotient:** $n^a \div m^a = (n \div m)^a$ $(m \neq 0)$.

Memory hook: multiplying powers → **add**; dividing powers → **subtract**; a power raised to a power → **multiply**.

Note: These laws also work for zero and negative exponents (next concept). Here we practise with positive exponents.

Where marks slip away

Trap 1 — Adding when you should multiply. $(5^2)^3$ is power-of-a-power, so multiply: 5^6. It is NOT $5^{2+3} = 5^5$.

Trap 2 — Multiplying the bases in the product law. $2^3 \times 2^5 = 2^8$, NOT 4^8. The base stays the same; only the exponents add.

Trap 3 — Using the laws with different bases. The add/subtract laws need the **same base**. $2^3 \times 3^3$ uses law 4: $(2\times3)^3 = 6^3$, not 6^6 or 5^3.

Trap 4 — Direction of subtraction. $n^a \div n^b = n^{a-b}$ (top minus bottom). Don't flip it.

Slip 5 — Leave it in exponential form unless asked for the value. $3^4 \times 3^2 = 3^6$ is a complete answer.

Model answers — write them like this

Example A. Simplify $3^4 \times 3^2$ and write in exponential form.

Step 1: Same base 3, multiplying → use the product law (add exponents).

Step 2: $3^4 \times 3^2 = 3^{4+2} = 3^6$.

$\boxed{\therefore\ 3^4 \times 3^2 = 3^6}$

Example B. Simplify $6^7 \div 6^3$.

Step 1: Same base 6, dividing → quotient law (subtract exponents).

Step 2: $6^7 \div 6^3 = 6^{7-3} = 6^4$.

$\boxed{\therefore\ 6^7 \div 6^3 = 6^4}$

Example C. Simplify $(5^2)^3$.

Step 1: A power raised to a power → multiply the exponents.

Step 2: $(5^2)^3 = 5^{2x3} = 5^6$.

$\boxed{\therefore\ (5^2)^3 = 5^6}$

Example D. Write 8^6 as a power of a power in two different ways.

Idea: Split the exponent 6 into two factors in different ways: $6 = 2 \times 3 = 3 \times 2$.

Way 1: $8^6 = (8^2)^3$.

Way 2: $8^6 = (8^3)^2$.

$\boxed{\therefore\ 8^6 = (8^2)^3 = (8^3)^2}$

Example E. Simplify $2^5 \times 2^3 \div 2^4$ and find its value.

Step 1: Same base — add then subtract exponents: $5 + 3 - 4 = 4$.

Step 2: So the expression $= 2^4 = 16$.

$\boxed{\therefore\ 2^5 \times 2^3 \div 2^4 = 2^4 = 16}$

Next: try the worksheet, then check the answer key.